Training the Next Generation of Curators at Genomes Online Database (GOLD)

Supratim Mukherjee, Galina Ovchinnikova, Dimitrios Stamatis, Cindy Li & T.B.K. Reddy

DOE Joint Genome Institute, Lawrence Berkeley National Laboratory, Berkeley, CA 94720

JGI JOINT GENOME INSTITUTE

Introduction

The Genomes Online Database (GOLD) is a manually curated online resource that catalogs metadata from sequencing projects from around the world. This metadata is collected, curated, organized and disseminated by GOLD, which is fully compliant with the Genomic Standards Consortium (GSC) Minimum Information about any (x) Sequence (MIxS) standards. Student interns are an integral part of GOLD which plays a pivotal role in developing the next generation of curators by providing hands-on experience and mentorship to undergraduate and high school students. Through its innovative training programs, GOLD connects students with real-world genomic data, allowing them to engage in curation tasks that are crucial to scientific research. By involving students in the curation process, GOLD fosters critical skills such as attention to detail, data analysis, and problem-solving, while also encouraging them to contribute to the scientific community through publications and presentations. These opportunities not only help students learn about metadata curation but also immerse them in the broader context of data management and database maintenance.

Where do GOLD's Interns Come From?

- Undergraduates from local community colleges and universities
- Highschool students - partner programs like BiotechPartners
- Students from Joint Genome Institute - UC Merced program
- Master's students from local universities

Biotech Partners education · careers · opportunities

UC Berkeley

DVC DIABLO VALLEY COLLEGE

JGI JOINT GENOME INSTITUTE A DOE OFFICE OF SCIENCE USER FACILITY

UNIVERSITY OF CALIFORNIA UCMERCED

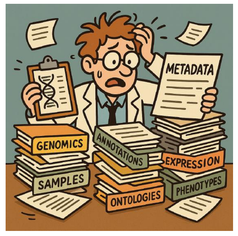
CAL STATE HAYWARD

What do they Work on & What Skills are Needed?

- A zeal to learn
- Knowledge of basic biology or related fields
- Computer skills like web browsing, using spread sheets
- Reading, comprehension and critical thinking skills
- Attention to detail

Why do we Need to Train Curators?

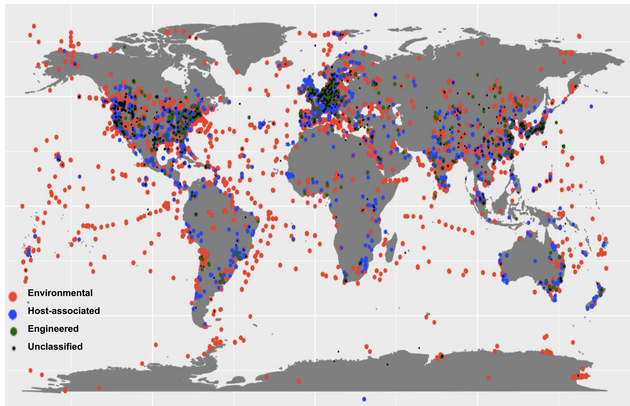
- Biocurators collate and organize data, knowledge and information in a structured format
- Majority of present-day curators come from wet lab and gain expertise in curation processes while working at various databases
- Introducing students to biocuration concepts early not only helps to develop future curators but also makes future researchers aware of the importance of metadata and devote enough resources to accurately document and share using standard protocols



What is Metadata and Why is it Important?

Metadata is information about information. For comparative analysis and hypothesis testing, it is extremely important to assure the quality of accurate metadata. GOLD has a growing list of over 1600 metadata fields spread across five broad entities.

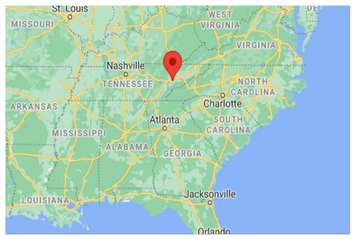
Category	Number of Projects	Number of Metadata Fields
Studies	62,193	26
Biosamples	217,100	720
Organisms	498,933	770
Sequencing Projects	588,518	60
Analysis Projects	438,258	53



Geographic Distribution of GOLD Biosamples & their Ecosystem Classification

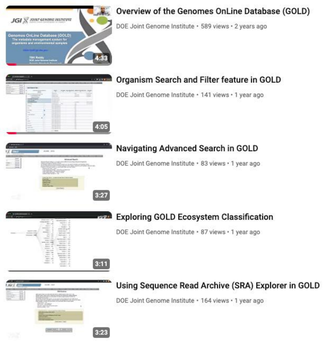
Student interns play a vital role in making sure that metadata is accurately represented in GOLD

35.9606, 83.9207 (*Incorrect longitude pointing to China*)



35.9606, - 83.9207 (*Correct longitude with negative value*)

GOLD's YouTube Page Powered by Interns



What Credits do GOLD Interns Receive?

JOURNAL ARTICLE

Twenty-five years of Genomes OnLine Database (GOLD): data updates and new features in v.9

Supratim Mukherjee, Dimitri Stamatis, Cindy Tianqing Li, Galina Ovchinnikova, Jon Bertsch, Jagadish Chandrabose Sundaramurthi, Mahathi Kandimalla, Paul A Nicolopoulos, Alessandro Favognano, I-Min A Chen, Nikos C Kyrpides, T B K Reddy

JOURNAL ARTICLE

Genomes OnLine database (GOLD) v.7: updates and new features

Supratim Mukherjee, Dimitri Stamatis, Jon Bertsch, Galina Ovchinnikova, Hema Y Katta, Alejandro Mojica, I-Min A Chen, Nikos C Kyrpides, TBK Reddy

JOURNAL ARTICLE

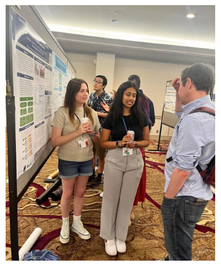
Genomes OnLine Database (GOLD) v.6: data updates and feature enhancements

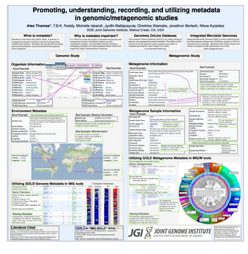
Supratim Mukherjee, Dimitri Stamatis, Jon Bertsch, Galina Ovchinnikova, Olena Verezemska, Michelle Isbandi, Alex D. Thomas, Rida Ali, Kaushal Sharma, Nikos C. Kyrpides, T. B. K. Reddy

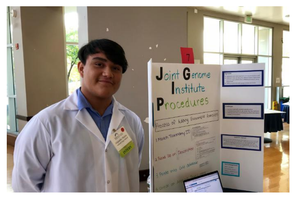
OCTOBER 31, 2016

Biotech Partners Summer Intern Co-Authors NAR Paper

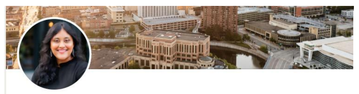
Kaushal Sharma, one of five Antioch High School rising juniors who participated in internships at the DOE Joint Genome Institute this summer through the Biotech Partners program is now an author on a *Nucleic Acids Research* publication. The paper describes the version 6 data updates and feature enhancements to the Genomes OnLine Database (GOLD) administered by the DOE JGI. T.B.K Reddy, Kaushal's mentor, described his student's contributions as reviewing close to 3,800 metagenomes to verify geographic locations and assign latitude and longitude values.







What are Some Past Interns Doing Now?



Mahathi Kandimalla (She/Her) · 2nd
Research @ Mayo Clinic | UC Berkeley '23
Rochester, Minnesota, United States · Contact info

Mayo Clinic
University of California, Berkeley

JGI **Research Intern**
Joint Genome Institute · Internship
Mar 2020 - Mar 2024 · 4 yrs 1 mo
Berkeley, California, United States

Worked on metadata curation for the GOLD database.

Rida Ali · 1st
Campus Representative at The Princeton Review

Student Research Assistant
Berkeley Lab
Dec 2015 - Sep 2017 · 1 yr 10 mos

Student Research Intern
Berkeley Lab
Jun 2015 - Sep 2015 · 4 mos

Faradia Pierre · 2nd
Family Medicine Physician
Jacksonville, Florida, United States · Contact info
484 connections

System Analyst
Lawrence Berkeley National Laboratory
Feb 2013 - Jun 2013 · 5 mos

- Organized metagenome metadata (internal or public databases, collaborators, etc).
- Architected controlled vocabularies and ontologies to improve the organization of metagenomic-related metadata.
- Interfaced with end-users and collaborators to gather feedback and improve the user support experience.

BioSciences

BERKELEY LAB

U.S. DEPARTMENT OF ENERGY | Office of Science

The work conducted by the Joint Genome Institute is supported by the Office of Science of the U.S. Department of Energy under Contract No. DE-AC02-05CH11231.